Introductory Organic Chemistry

Dr Eric Amigues

XJTLU 2024/2025

Semester 1

Introductory Organic Chemistry

About the module

- 2.5 credits module
- Lectures-2 hours per week
- Workshops/tutorials-1 hour per week

About the Assessment

- Final examinations-60%
- Class Test-20% (4th of November)
- Homework-10% (2 CW: 8th of October and 26th of November)
- In-class quizzes-10% (at anytime during lectures or tutorials)

Dr Eric Amigues

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There are no silly questions, only silly answers !!!!

Visiting hours:

Appointment ONLY during the week (9am-5pm)

Questions by e-mail: anytime

Try to think in English

(Chinese is prohibited during class hours)

Schedule

Date	Type of Lesson
9th of September	Lecture 1
10th of September	Tutorial 1
16th of September	Lecture 2
17th of September	No Class Mid-autumn festival
23rd of September	Lecture 3
24th of September	Tutorial 2
7th of October	Lecture 4
8th of October	Tutorial 3
14th of October	Lecture 5

Schedule

- **Assignment of Homework 1 the day of Lecture1, of HW2 the day of lecture2 etc.**
- **One week to prepare homework (HW)**
- **HW is formative (no marks), there will be 2 courseworks**

Learning Outcomes

- **A draw structures and predict shapes of major classes of organic compounds**
- **B understand the principles of bonding in major classes of organic compounds**
- **C understand the basic principles of stereochemistry assignment**
- **D understand some important reactions of a range of functional groups**
- **E show an understanding of the major types of reaction mechanisms**

Syllabus

1. Organic structures
2. Structure of molecules
3. Organic reactions
4. Nucleophilic addition to the carbonyl group
5. Delocalization and conjugation
6. Acidity, basicity, and p*K*a
7. Using organometallic reagents to make C-C bonds
8. Nucleophilic substitution at the carbonyl (C=O) group
9. Nucleophilic substitution at C=O with loss of carbonyl oxygen
10. Stereochemistry

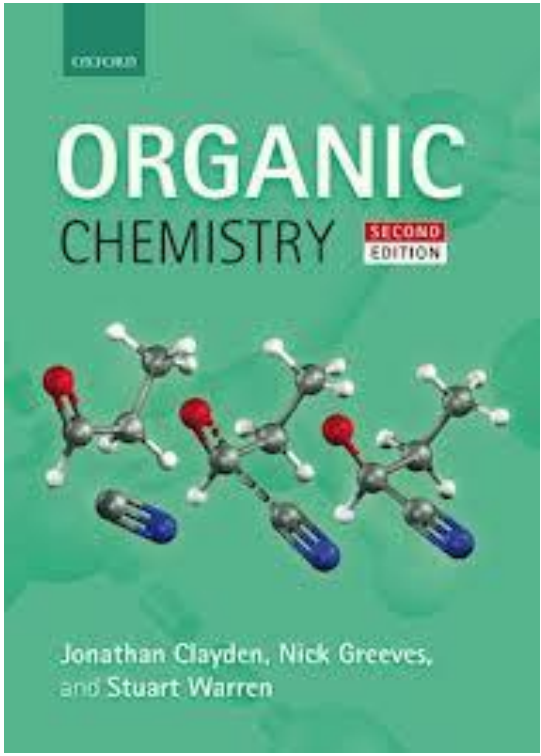
Recommended Textbooks

Organic Chemistry

J. Clayden, N. Greeves, S. Warren, P.

Oxford University Press, 2001

Chapters 2, 4-11, 14-17, 19-21



Reference Books: Any Organic Chemistry books available to you, you can find many in the library

Also: The book of General Chemistry of Housecroft and Constable

What is Organic Chemistry?

Organic Chemistry: The Chemistry of carbon-based compounds

18th century:

- **Compounds from lifeless matter – *Inorganic***
- **Compounds from living organisms – *Organic***
- **Organic compounds thought to have "vital force"**

Friedrich Wöhler



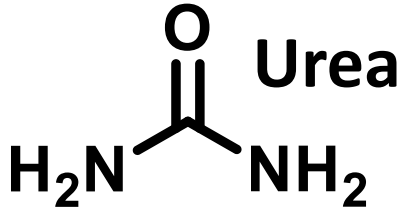
Ammonium cyanate

NH_4OCN

(mineral sources)

Inorganic





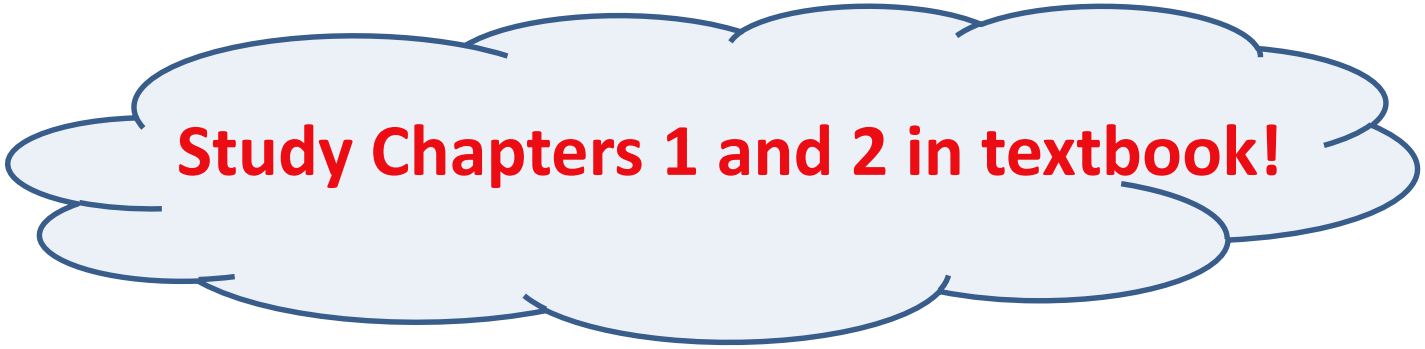
Organic

Studying Organic Chemistry

- **Structure determination**: How to find out the structures of organic compounds
- **Theoretical Organic Chemistry**: How to understand those structures in terms of atoms and electrons that bind them together
- **Reaction Mechanisms**: How to find out how molecules react and how to predict their reactions
- **Synthesis**: How to design new molecules, and then make them
- **Biological Chemistry**: How to find out what Nature does and how the structures of biologically active molecules are related to what they do

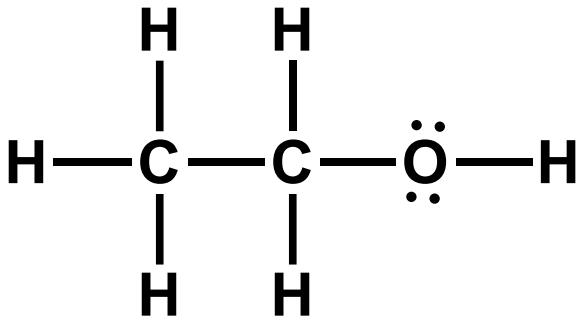
Organic Structures

- Why we use particular diagrams
- How chemists name molecules in writing and in speech?
- What is the skeleton of an organic molecule?
- What is a functional group?
- Some abbreviations used by all (organic) chemists
- Drawing organic molecules realistically in an easily understood style

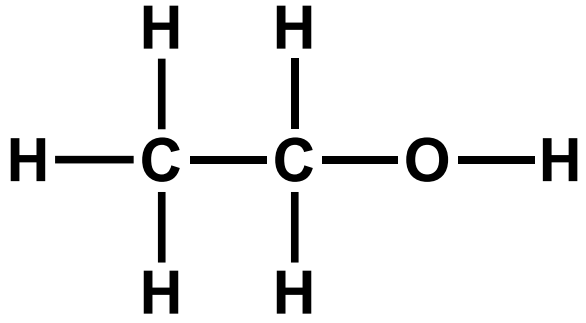


Lewis Structures-Representation of Ethanol C_2H_6O

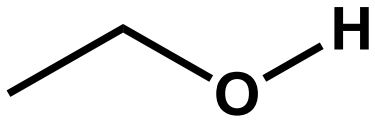
Each shared electron pair-a bond between two atoms-is drawn as a straight line



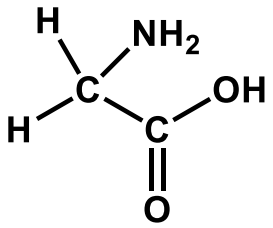
Lone pairs are often not shown (but are implied)



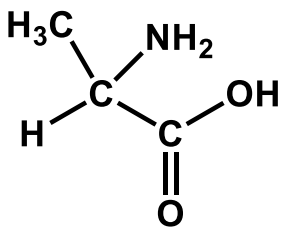
Hydrogens are often not shown (but are implied)



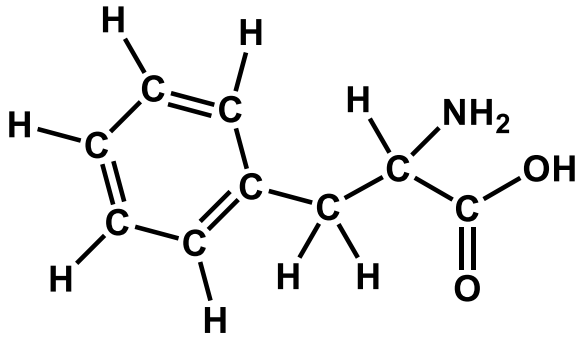
Too Detailed Organic Structures



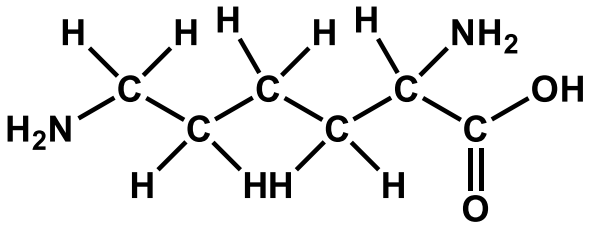
glycine



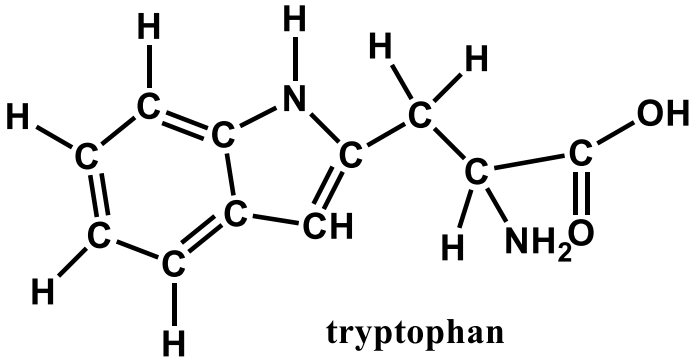
alanine



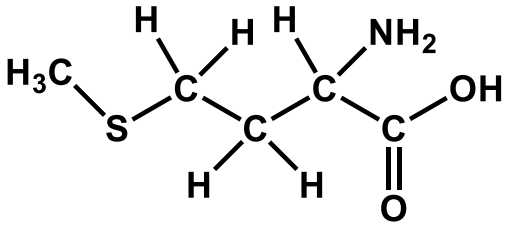
phenylalanine



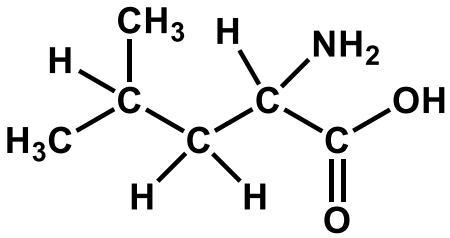
lysine



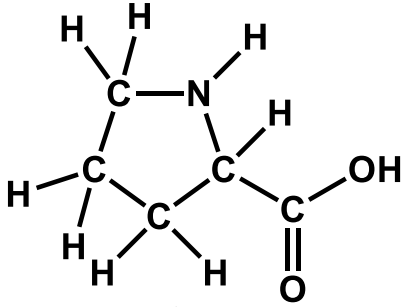
tryptophan



methionine



leucine



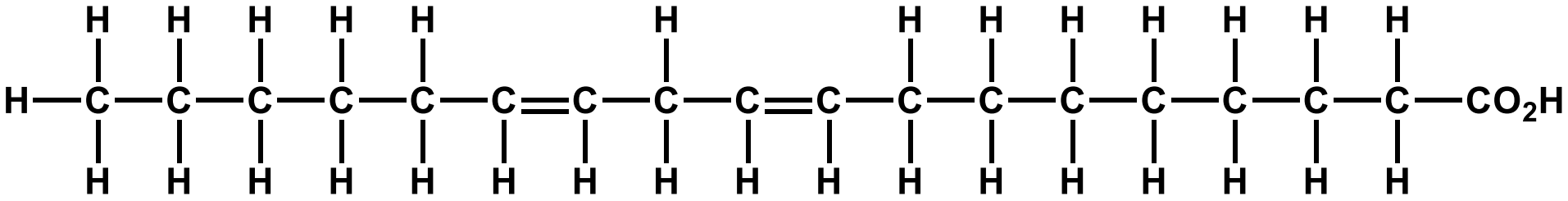
proline

Useless Organic Structures

$CH_3CH_2CH_2CH_2CH_2CH{=}CHCH_2CH{=}CHCH_2CH_2CH_2CH_2CH_2CH_2CH_2CO_2H$

linoleic acid

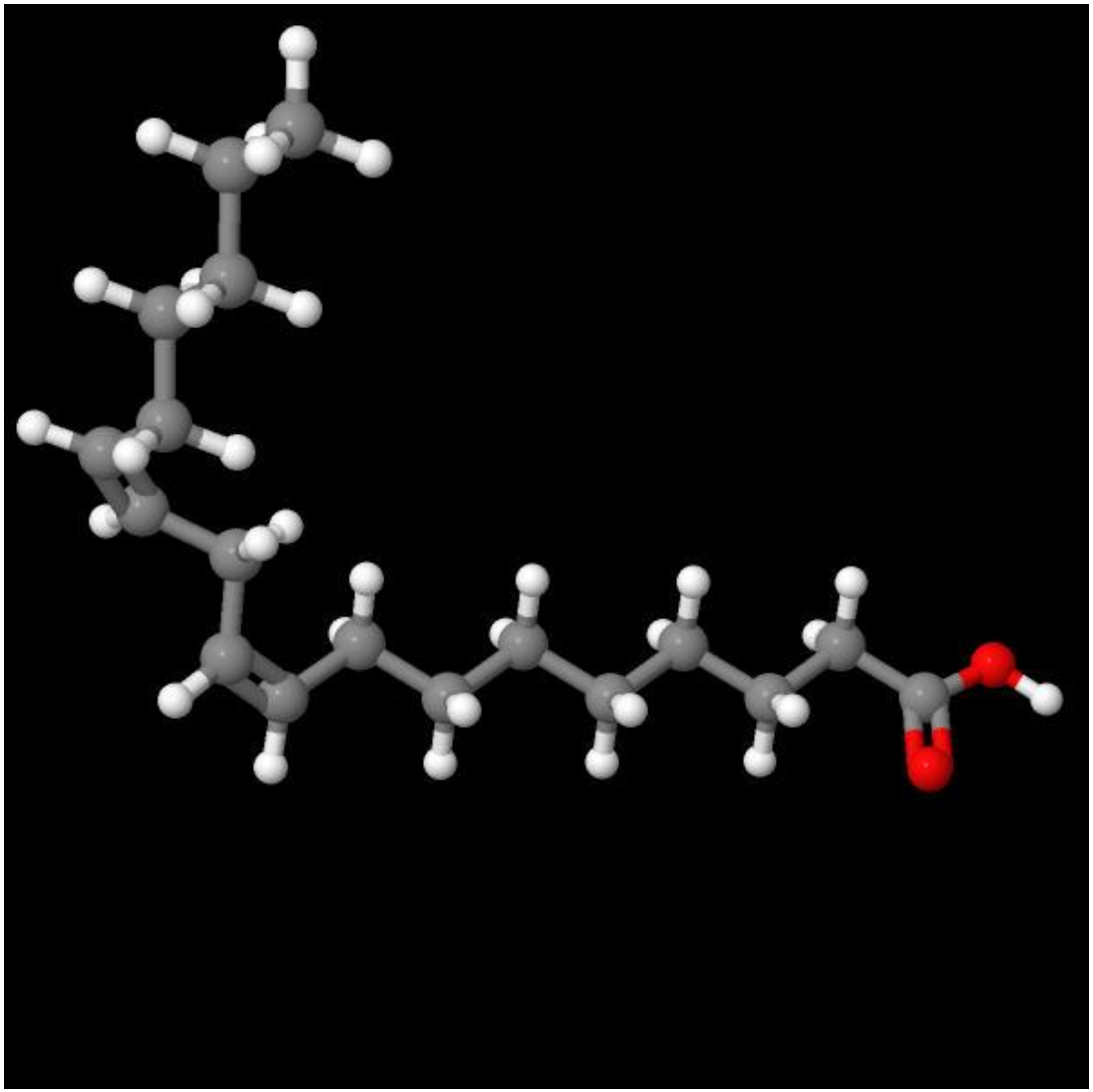
Linear, hides the structure and functional groups



linoleic acid

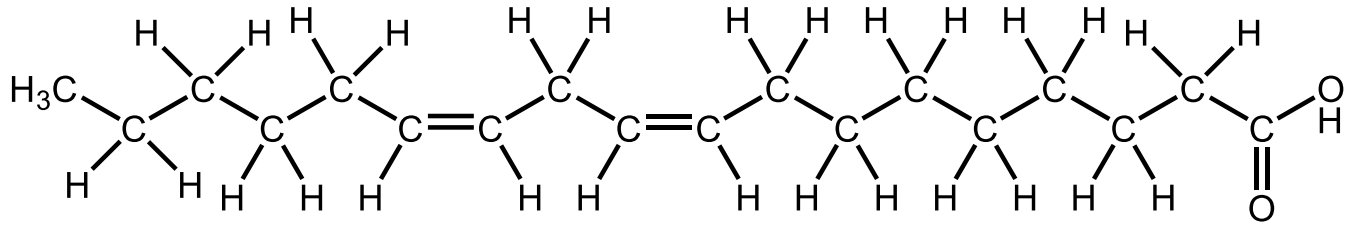
Unrealistic 90° angles, all those hydrogens!

Linoleic acid Structure

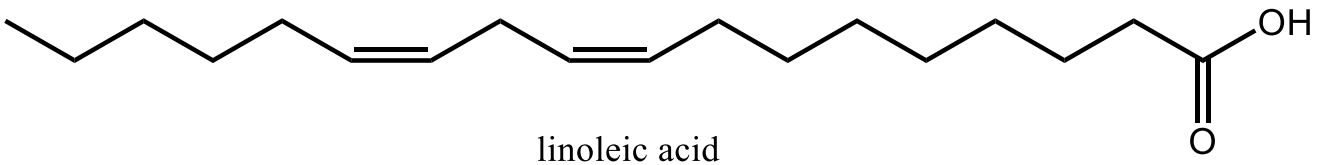


For interactive resources visit http://www.chemtube3d.com/ClaydenLinoleic.html

Better Organic Structures



linoleic acid



linoleic acid

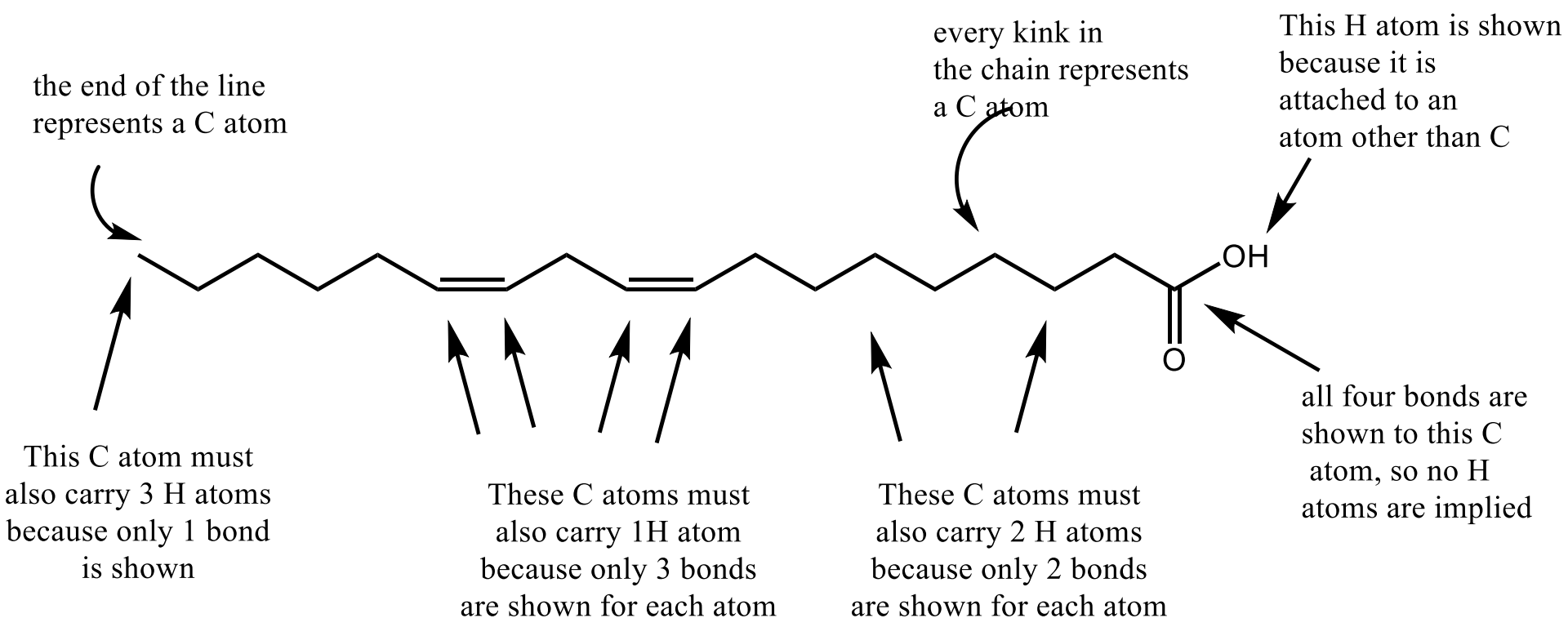
Guideline 1: Draw chains of atoms as zig-zags

Guideline 2: Miss out the H's attached to carbon atoms, along with the C-H bonds (unless there is a good reason not to)

Guideline 3: Miss out the capital C's representing carbon atoms (unless there is a good reason not to)

Guideline 4: Always bear in mind that C's form 4 bonds

Ideal Organic Structures



Guidelines

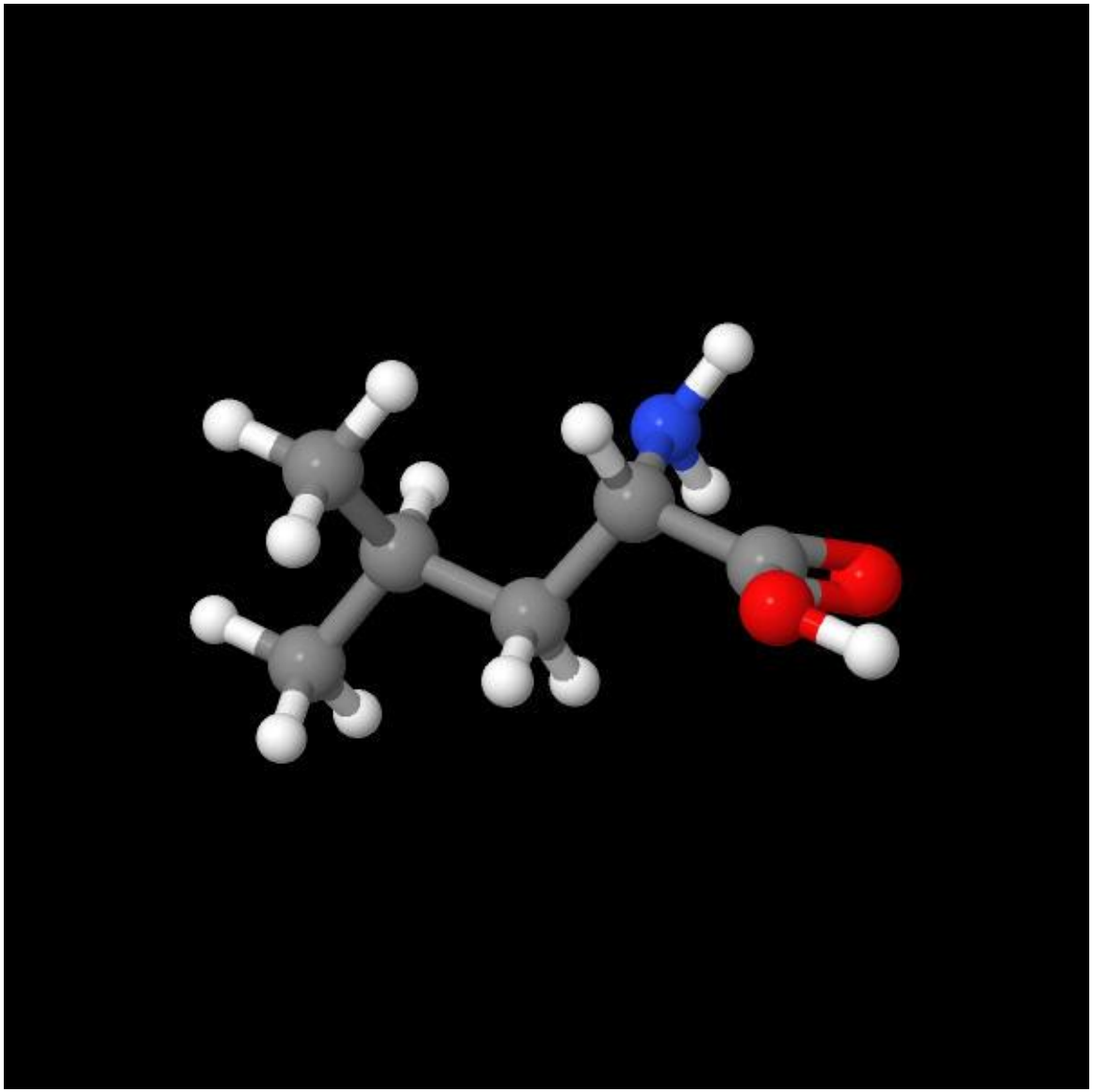
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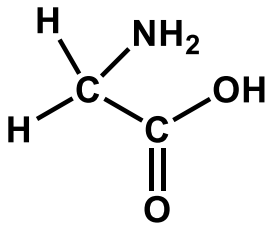
Guideline 4: Always bear in mind that C's form 4 bonds and also that double bonds are planar (120°) and triple bonds linear (180°)

Leucine Structure

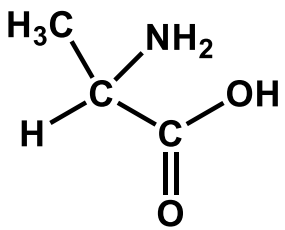


Visit http://www.chemtube3d.com/ClaydenAminoAcids.html

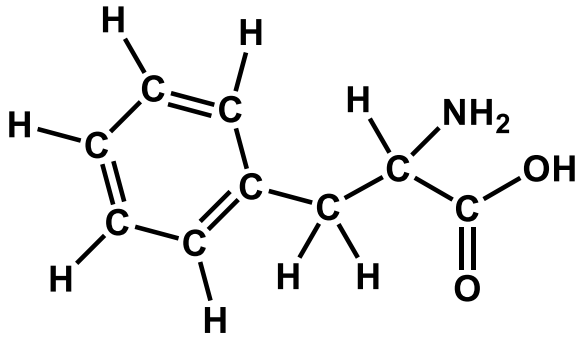
Simplify these too detailed structures!



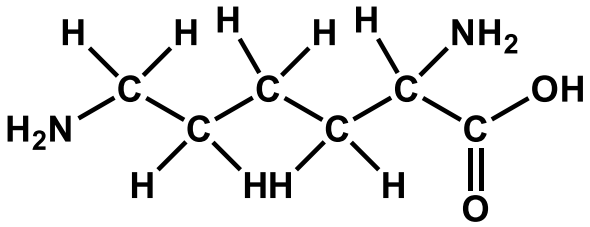
glycine



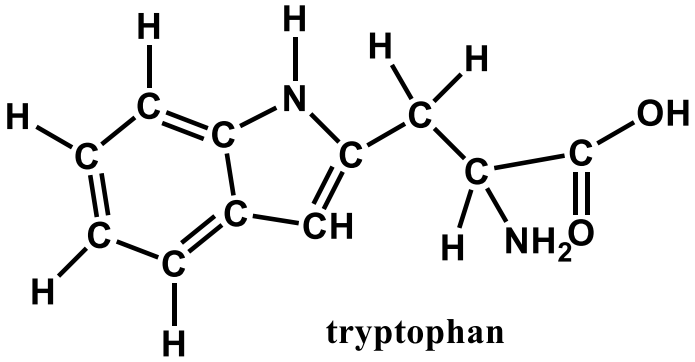
alanine



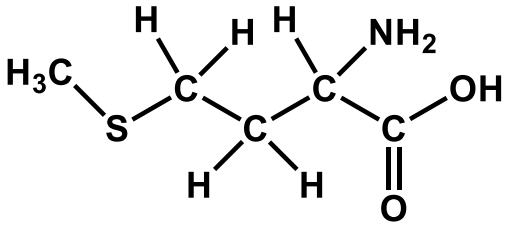
phenylalanine



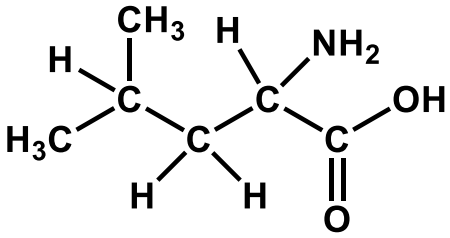
lysine



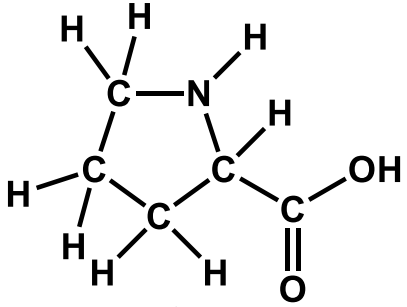
tryptophan



methionine

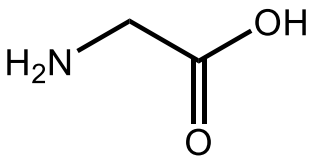


leucine

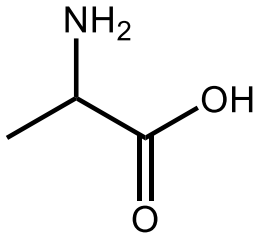


proline

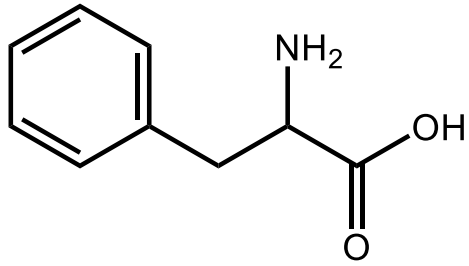
Much clearer Organic Structures



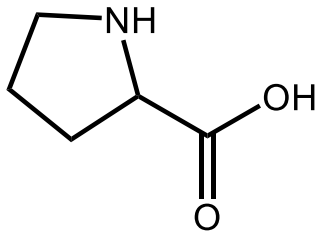
glycine



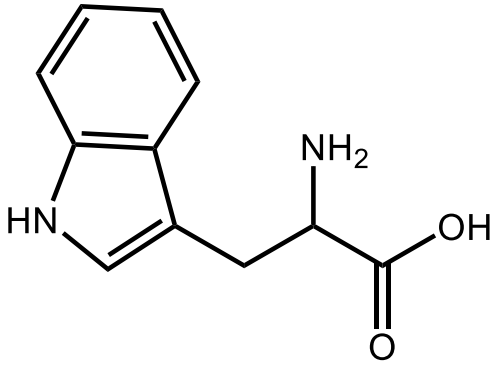
alanine



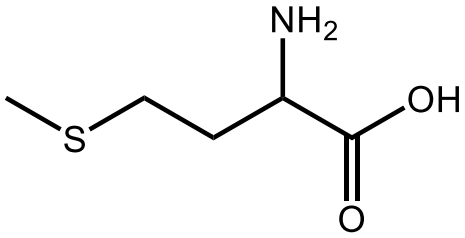
phenylalanine



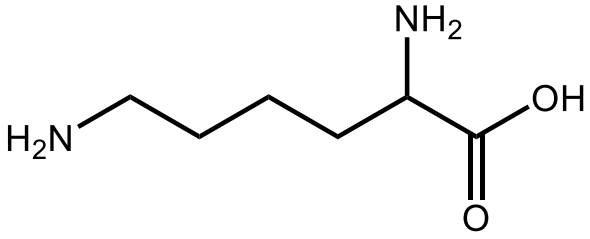
proline



tryptophan



methionine

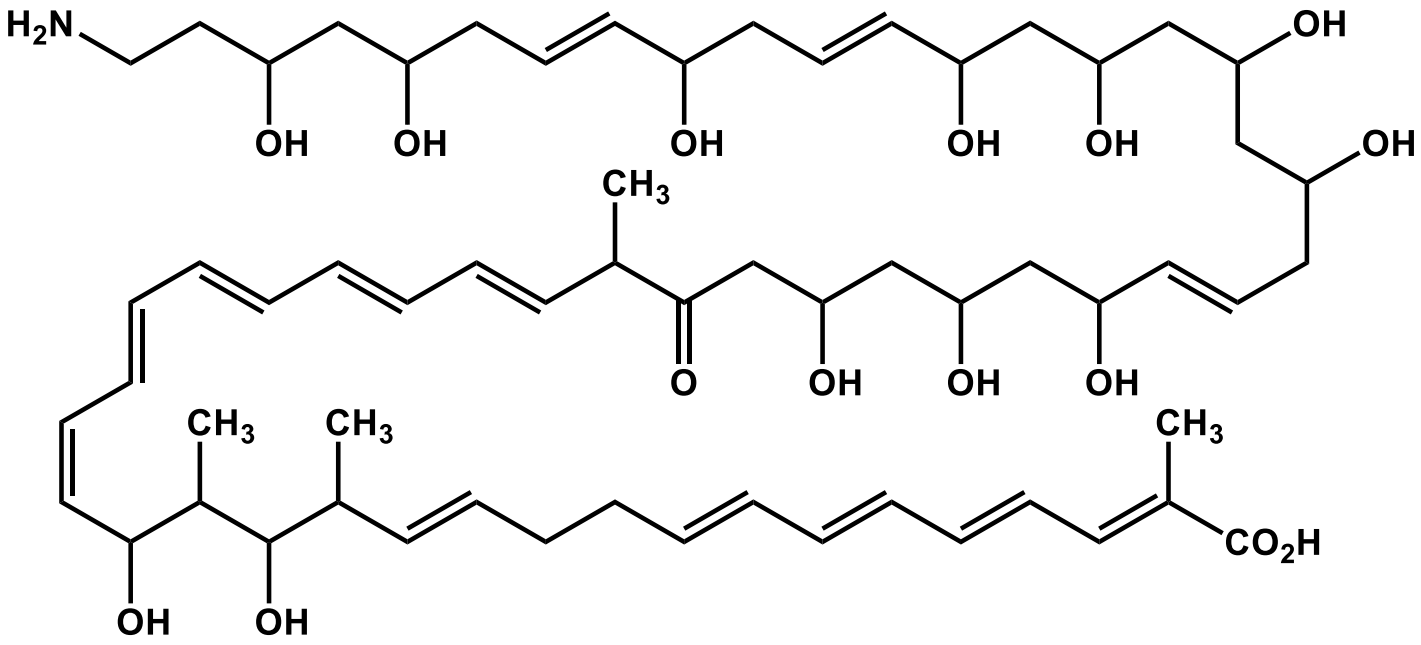


lysine

Organic Structures-Naming Organic Compounds

- **Names for Chains, Rings, Branches: What is the skeleton (hydrocarbon framework) of an organic molecule?**
- **Functional Groups**
- **Small structures to learn**
- **Abbreviations used by all organic chemists**

Chains of carbon atoms



linearmycin

Linearmycin: 57-carbon linear chain!

Names for Chains of carbon atoms

Names and abbreviations for carbon chains

Number of carbon atoms in chain	Name of group	Formula†	Abbreviation	Name of alkane (= chain + H)
1	methyl	$-CH_3$	Me	methane
2	ethyl	$-CH_2CH_3$	Et	ethane
3	propyl	$-CH_2CH_2CH_3$	Pr	propane
4	butyl	$-(CH_2)_3CH_3$	Bu	butane
5	pentyl	$-(CH_2)_4CH_3$	—‡	pentane
6	hexyl	$-(CH_2)_5CH_3$	—‡	hexane
7	heptyl	$-(CH_2)_6CH_3$	—‡	heptane
8	octyl	$-(CH_2)_7CH_3$	—‡	octane
9	nonyl	$-(CH_2)_8CH_3$	—‡	nonane
10	decyl	$-(CH_2)_9CH_3$	—‡	decane

† This representation is not recommended.

‡ Names for longer chains are not commonly abbreviated.

IUPAC Systematic nomenclature: Replacing the ending –ane in the alkane by –yl

IUPAC: International Union of Pure and Applied Chemistry

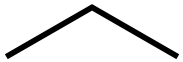
Names for alkanes-Examples



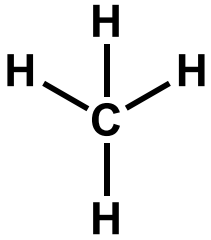
pentane



octane



propane

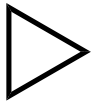


methane

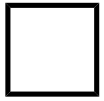
Alkane=Paraffin, derived from Latin parum+affinis=lacking affinity, ie reactivity

Names for Rings of carbon atoms

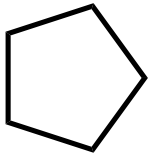
Aliphatic



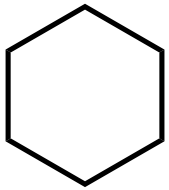
cyclopropane



cyclobutane

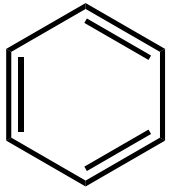


cyclopentane

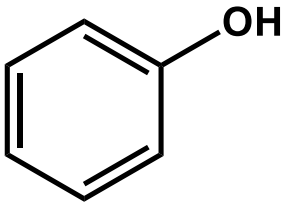


cyclohexane

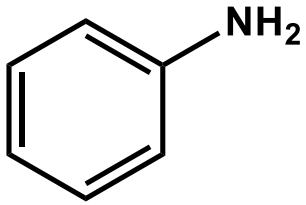
Aromatic



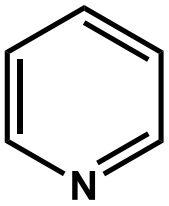
benzene



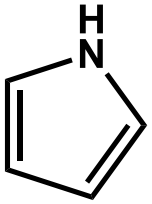
Phenol



aniline



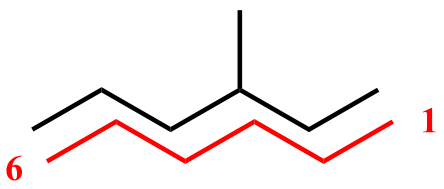
pyridine



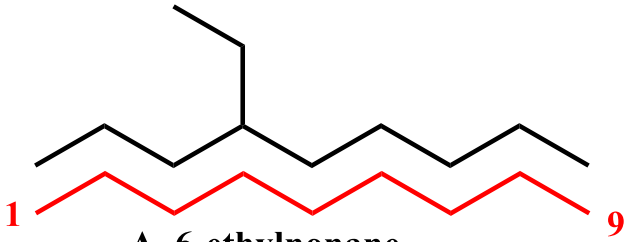
pyrrole

Names for Branches-Simple Examples

- Look at the longest continuous chain, count the C's
- Find the name of the branch
- Compose the name of the molecule, the number indicating the branch should be the lowest possible
- What is the name of these molecules?



A. 2-ethylpentane
B. 4-methyhexane
C. 3-methylhexane



A. 6-ethylnonane
B. 3-propyloctane
C. 4-ethylnonane

How to represent properly organic structures: The guidelines

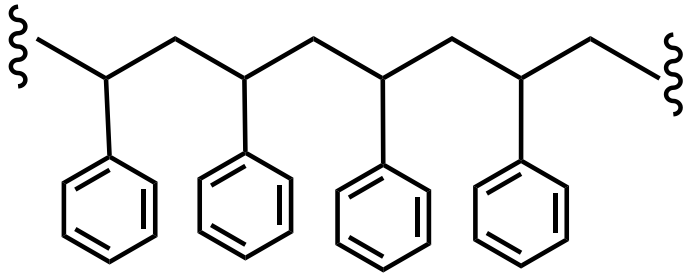
Guideline 1: Draw chains of atoms as zig-zags

Guideline 2: Miss out the H's attached to carbon atoms, along with the C-H bonds (unless there is a good reason not to)

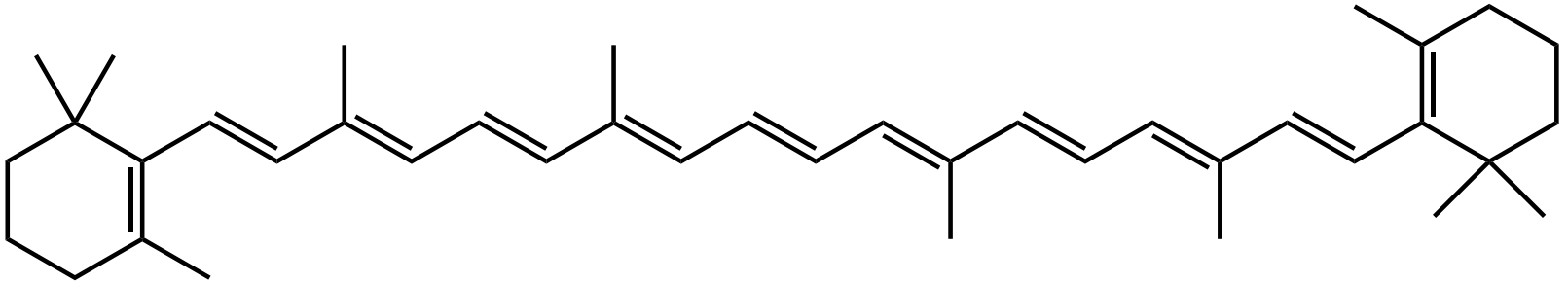
Guideline 3: Miss out the capital C's representing carbon atoms (unless there is a good reason not to)

Guideline 4: Always bear in mind that C's form 4 bonds and also that double bonds are planar (120°) and triple bonds linear (180°)

Branches

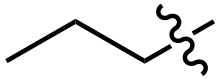


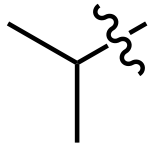
Part of the structure of polystyrene



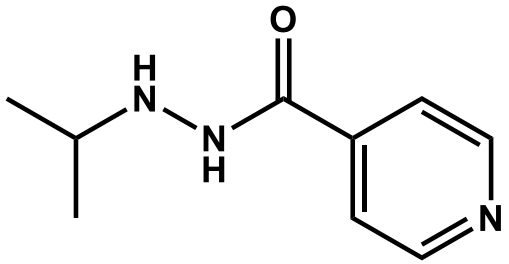
β-carotene

Branched Branches: isopropyl group

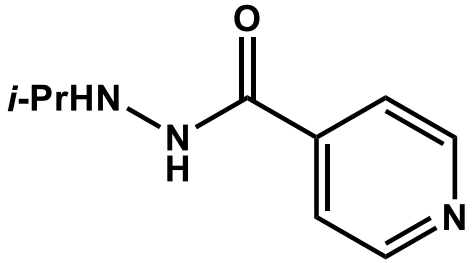




n-Propyl is a straight chain, *i*-Propyl is Branched



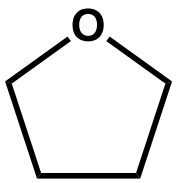
is equivalent to

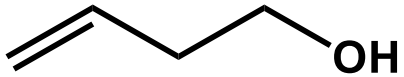


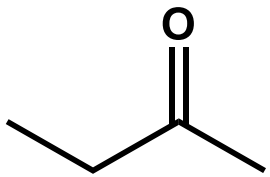
iproniazid
(antidepressant)

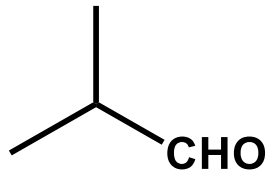
Isomers-Definition

- **Isomers are molecules with the same kinds and numbers of atoms joined up in different ways n-propanol (*n*-PrOH) and isopropanol (*i*-PrOH), are isomeric alcohols.**
- **Isomers need not have the same functional group, these compounds are all isomers of C_4H_8O.**



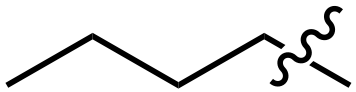




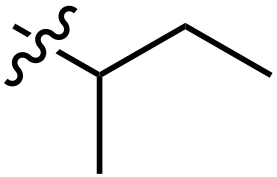


Isomers: molecules with the same molecular formula

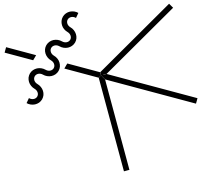
Branched Branches: Butyl chains
normal, iso, sec, tert



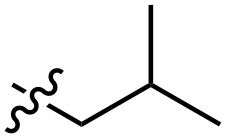
The *n*-Butyl group
n-Bu



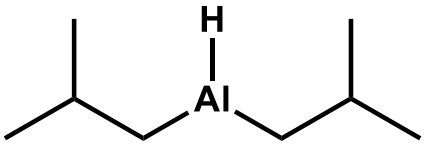
The *sec*-Butyl group
s-Bu



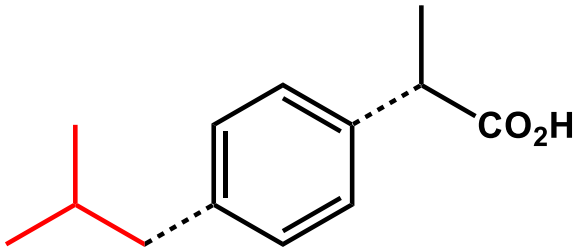
The *tert*-Butyl group
t-Bu



The *iso*-Butyl group
i-Bu



diisobutyl aluminium hydride (DIBAL)
is equivalent to HAl*i*-Bu_2

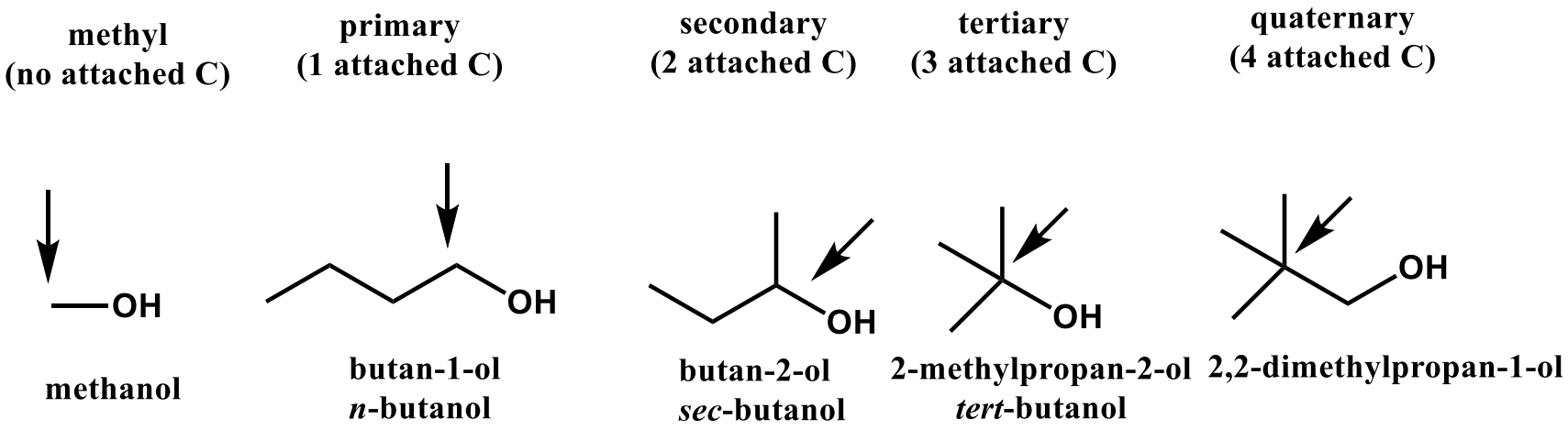


Ibuprofen

Useful Descriptors

Primary, secondary, and tertiary

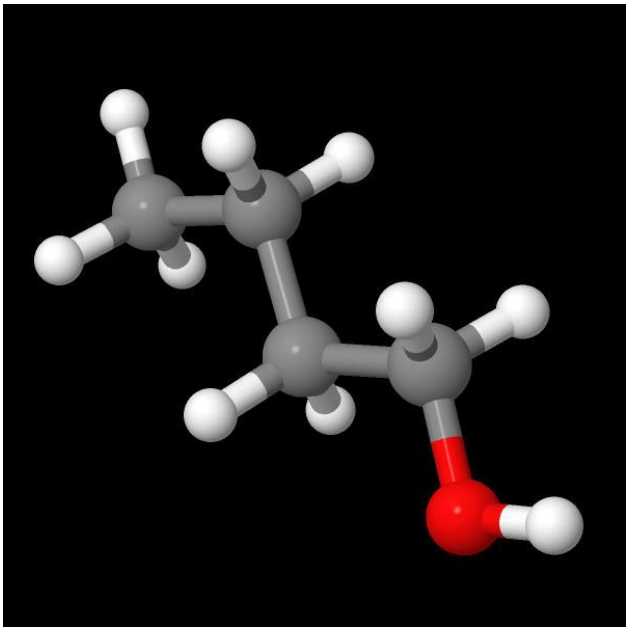
The prefixes *sec* and *tert* are really short for secondary and tertiary, terms that refer to the carbon atom that attaches these groups to the rest of the molecular structure

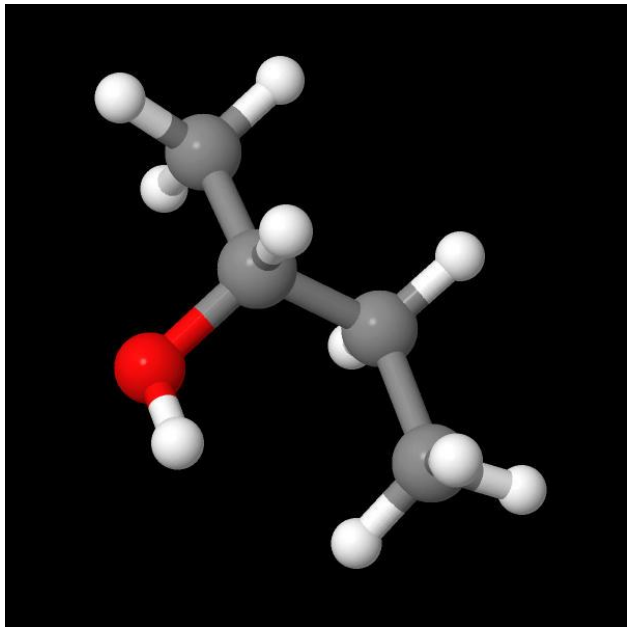


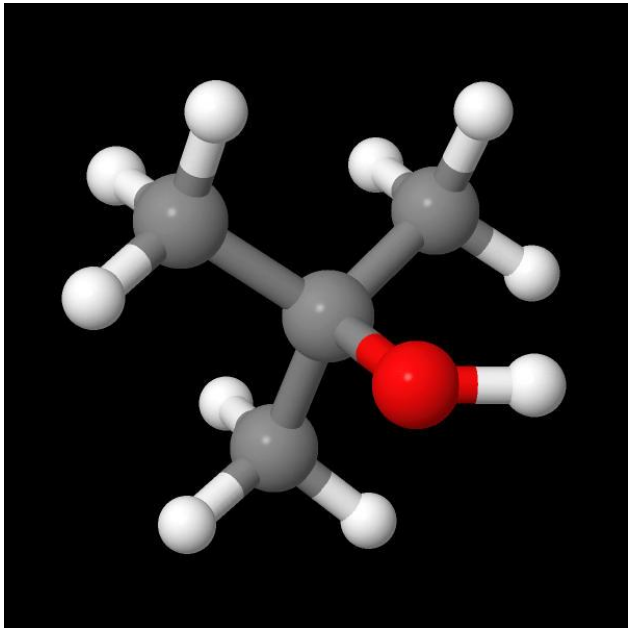
A primary carbon is attached to only one C atom, a secondary to two other C atoms, and so on. This means there are five types of carbon atom.

These names for bits of hydrocarbon framework are more than just useful ways of writing or talking about chemistry. They tell us something fundamental about the molecule and we shall use them when we describe reactions

Structure of Butanols







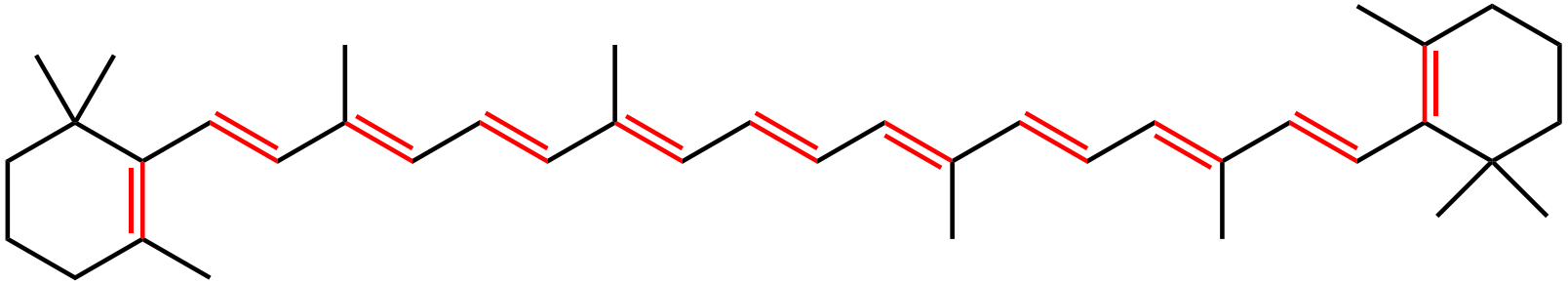
Organic Structures- Functional Groups

- **Names for Chains, Rings, Branches: What is the skeleton (hydrocarbon framework) of an organic molecule**
- **Functional Groups**
- **Small structures to learn**
- **Abbreviations used by all organic chemists**

Functional Groups-Hydrocarbons

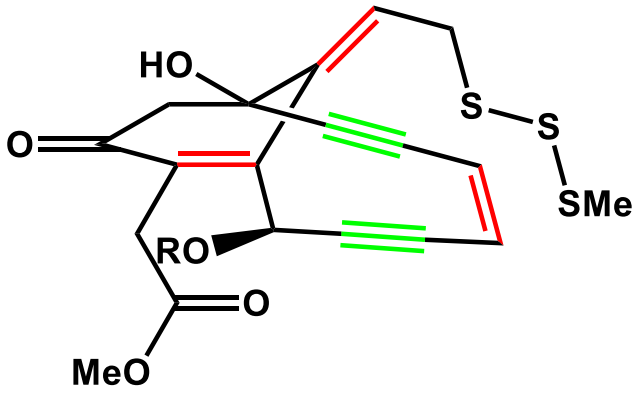
Alkanes contain no functional groups

Alkenes (olefins) contain C=C double bonds

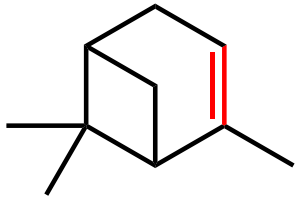


β-carotene

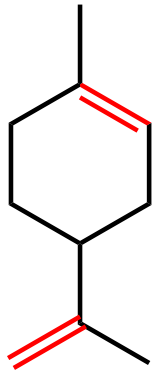
Alkynes contain C≡C triple bonds



calicheamicin
(R= a string of sugar molecules)



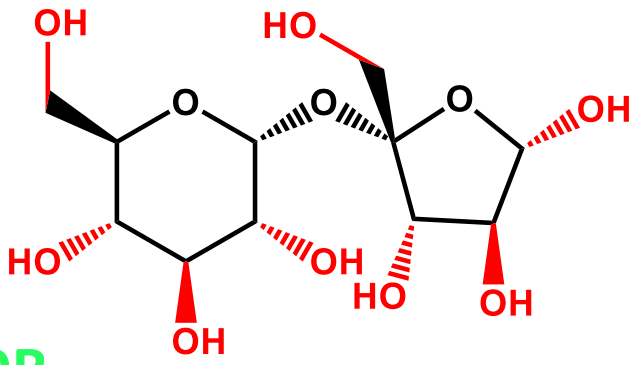
α-pinene



limonene

Functional Groups-Alcohols and Ethers

Alcohols R-OH contain a hydroxyl group OH

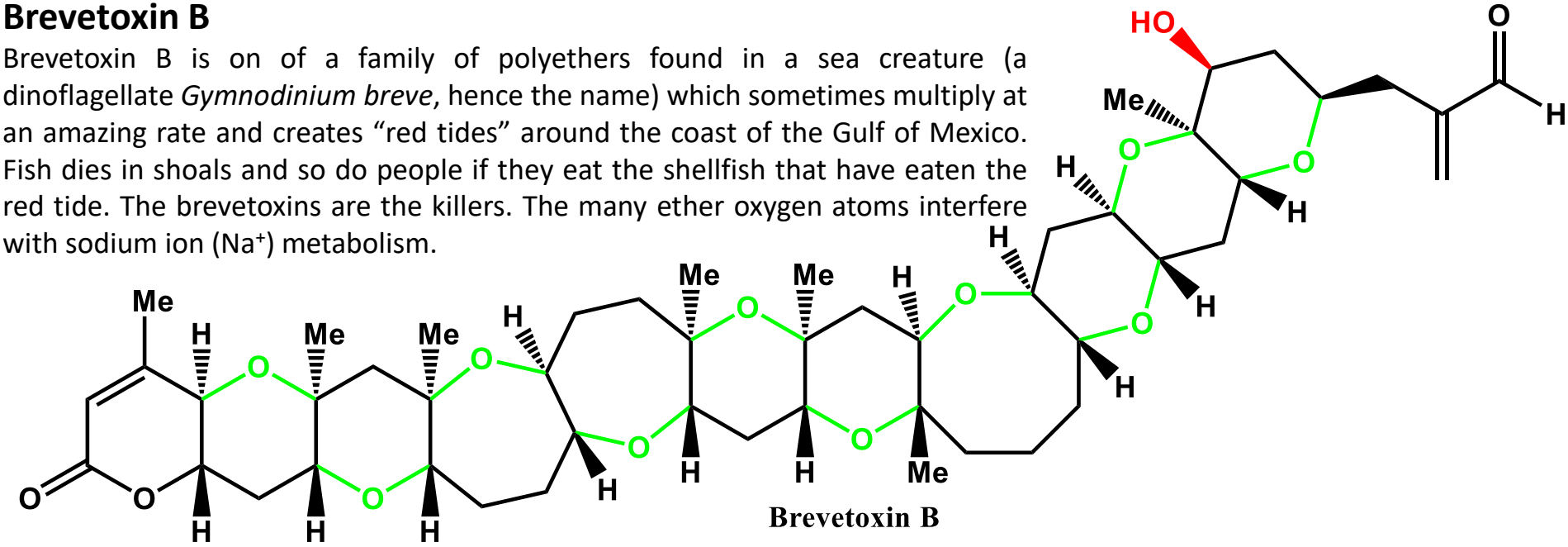


sucrose = sugar

Ethers R^1-O-R^2 contain an alkoxy group OR

Brevetoxin B

Brevetoxin B is on of a family of polyethers found in a sea creature (a dinoflagellate *Gymnodinium breve*, hence the name) which sometimes multiply at an amazing rate and creates "red tides" around the coast of the Gulf of Mexico. Fish dies in shoals and so do people if they eat the shellfish that have eaten the red tide. The brevetoxins are the killers. The many ether oxygen atoms interfere with sodium ion (Na^+) metabolism.

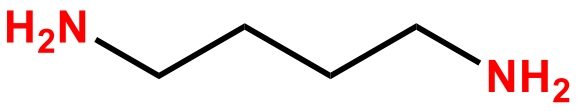


Brevetoxin B

Functional Groups-Amines and Alkylhalides

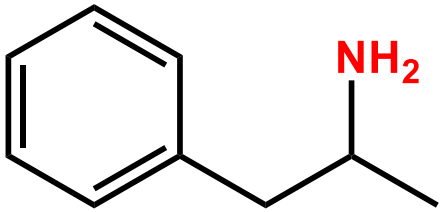
Amines R–NH_2 contain the amino group NH_2

Formed as meat decays



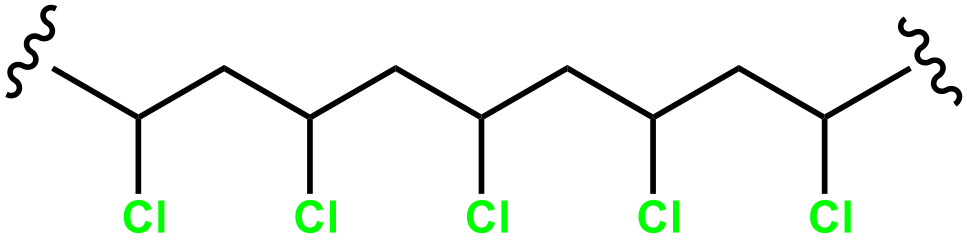
putrescine

Notorious stimulant



amphetamine

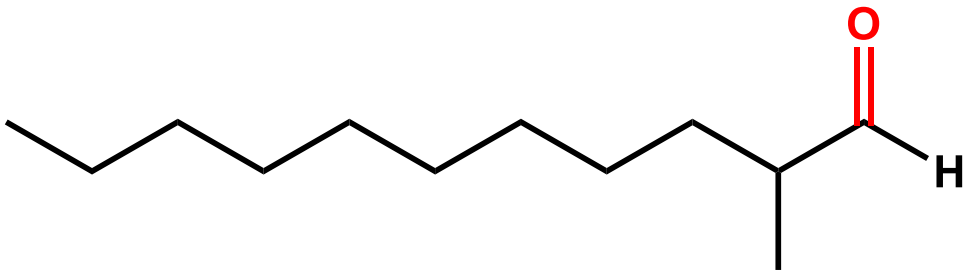
Alkylhalides R–X contain halogens, X can be F, Cl, Br, I



a section of the structure of PVC

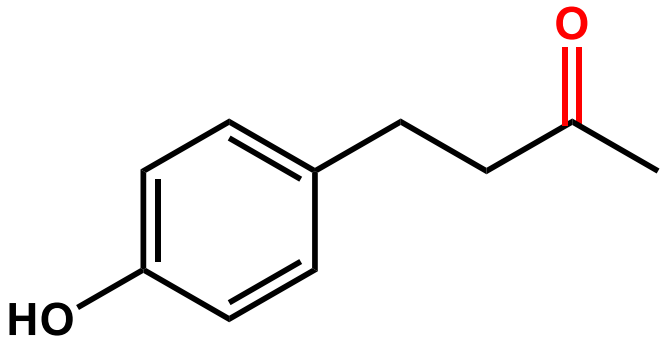
Functional Groups-Aldehydes and Ketones

Aldehydes R–CHO and ketones R^1–CO–R^2 contain the C=O carbonyl group



2-methylundecanal

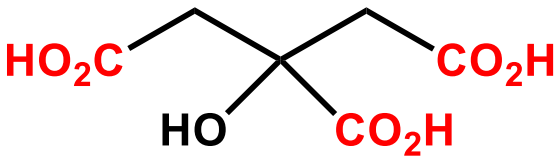
Chanel Nº 5



"raspberry ketone"

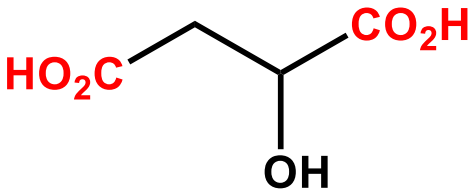
Functional Groups-Carboxylic acids and Derivatives

Carboxylic acids R–CO_2H contain the carboxyl group CO_2H



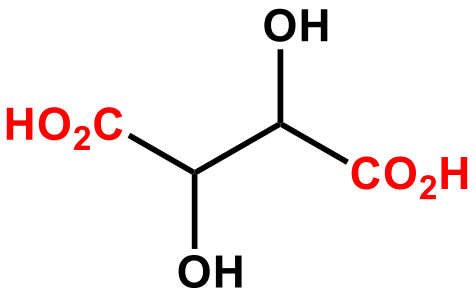
citric acid

Found in lemons



malic acid

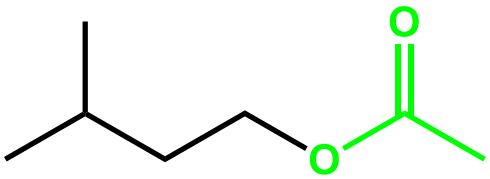
…in apples



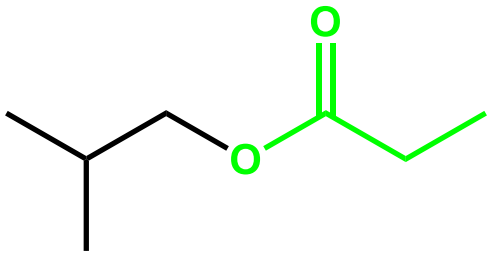
tartaric acid

…in grapes

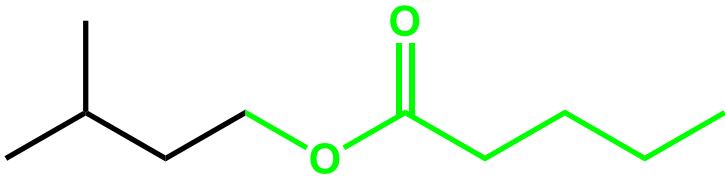
Esters R^1–CO_2R^2 contain a carboxyl group with an extra alkyl group CO_2R^2



isopentyl acetate
(bananas)



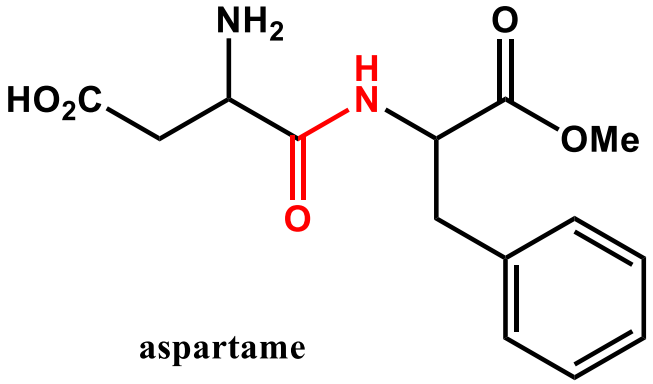
isobutyl propionate
(rum)



isopentyl valerate
(apples)

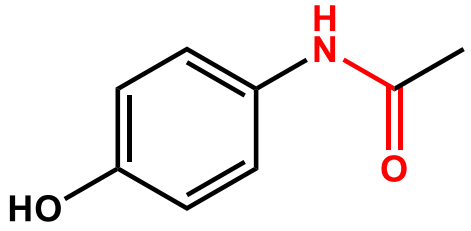
Functional Groups-Derivatives of Carboxylic acids

Amides R–CONHR contain N-C=O



aspartame

Artificial sweetener

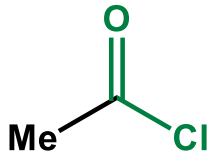


paracetamol

Pain-killer

Acylchlorides R–COCl contain the group COCl

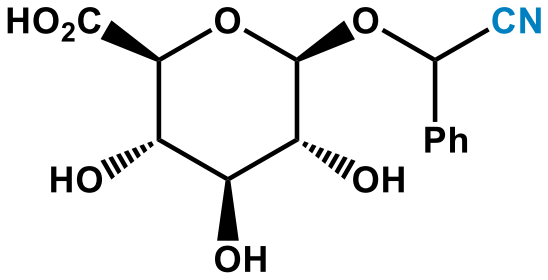
Acetylating agent



acetyl chloride

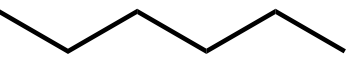
Nitriles or **cyanides** R-CN contain the cyano group C≡N

Anti-cancer drug

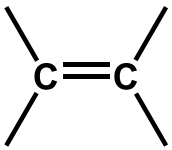


laetrile

Common Functional Groups



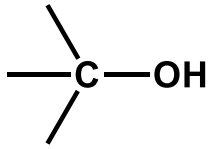
alkane



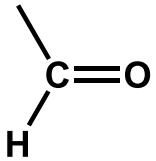
alkene



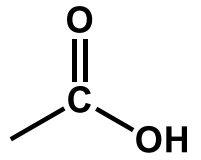
alkyne



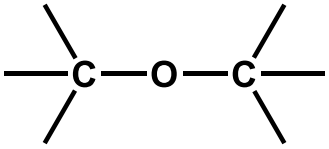
alcohol (1°, 2°, 3°)



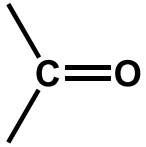
aldehyde



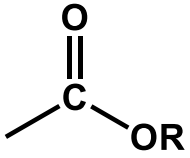
carboxylic acid



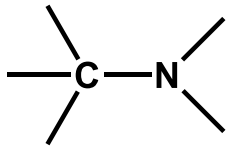
ether



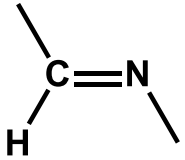
ketone



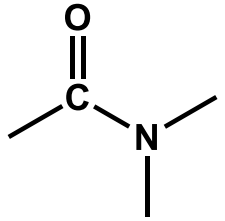
ester



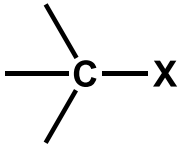
amine (1°, 2°, 3°)



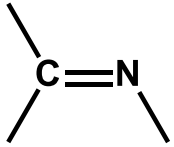
aldimine



amide (1°, 2°, 3°)



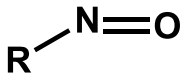
alkyl halide (1°, 2°, 3°)



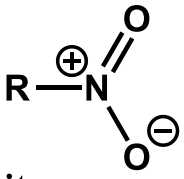
ketimine



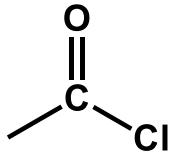
nitrile



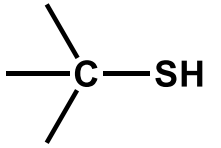
nitroso



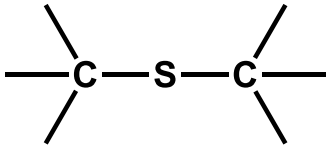
nitro compound



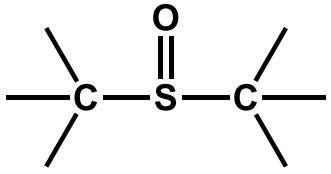
acyl chloride



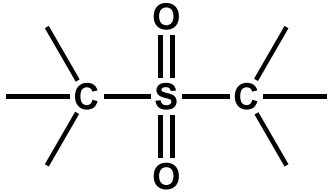
mercaptan, thiol



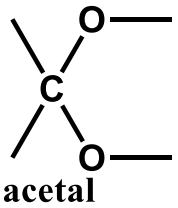
disulfide



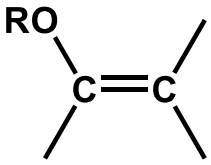
sulfoxide



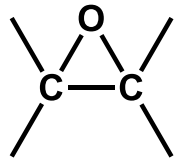
Sulfone



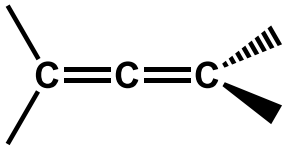
acetal



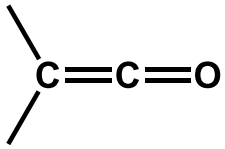
enol ether



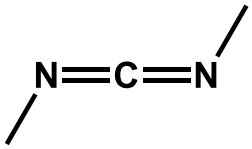
epoxide



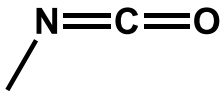
Allene



ketene



carbodiimide



isocyanate

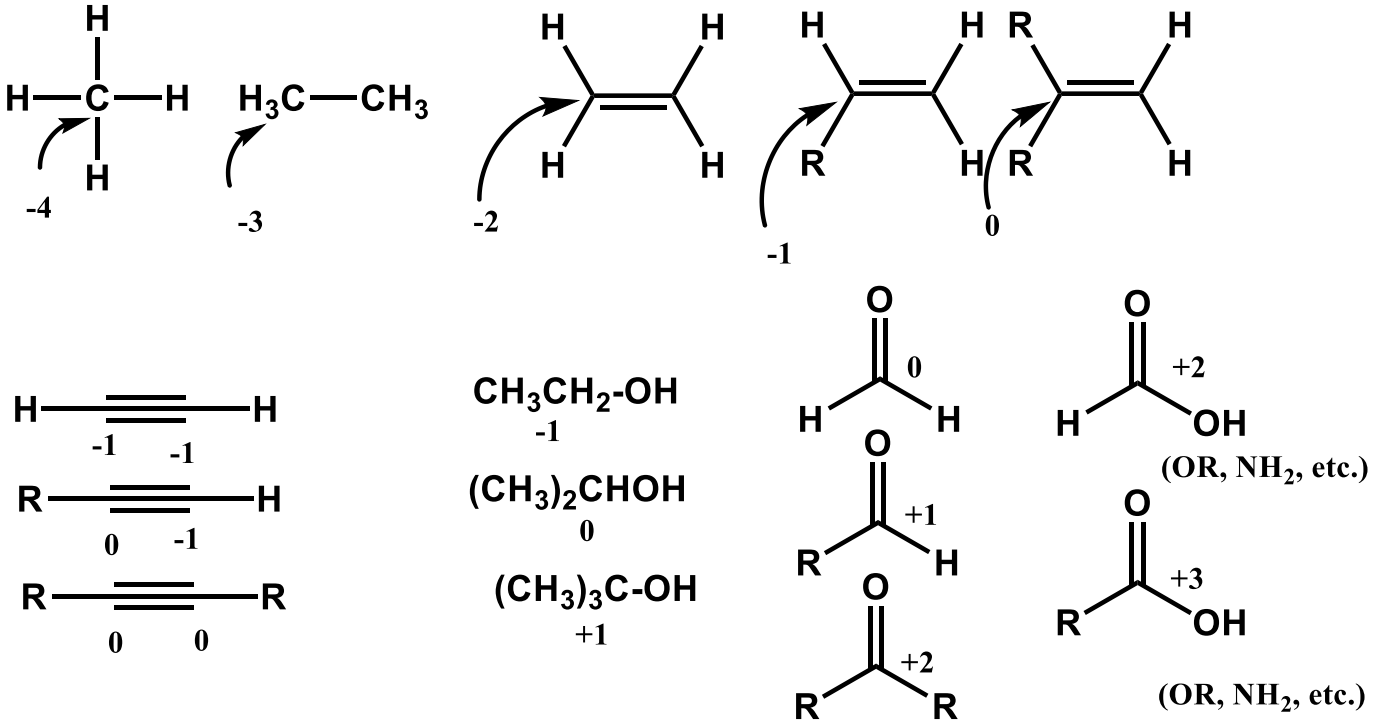
Important functional groups and oxidations levels

- **Oxidation is defined as the loss of electrons**
  - **Very straightforward when dealing with metal ions. $Mg^{0+} \rightarrow Mg^{2+}$ is an oxidation**
- **Reduction is defined as the gain of electrons.**
  - **The conversion of $Ag^{+} \rightarrow Ag^{0}$ involves the gain of one electron by the silver ion and thus silver is reduced.**
- **Because organic compound are mostly covalently bound, it is not as straightforward to determine if oxidation or reduction took place.**
- **A set of simple rules allow to determine carbon oxidation level.**

Oxidation levels Rules

- **Bonds to hydrogen or other elements more electropositive than carbon contribute to -1 to the oxidation level.**
- **Bonds to other carbon atoms contribute 0 to the oxidation level.**
- **Bonds to oxygen or other elements more electronegative than carbon contribute +1 to the oxidation level.**
- **Multiple bounds to an element count as multiple single bonds to that element.**
- **A pair of electrons on carbon contribute -1 to the oxidation level.**
- **A positive charge on carbon contribute +1 to the oxidation level**

Oxidation levels



Summary

- **Why we use particular diagrams?**
- **How organic chemists name molecules in writing and in speech?**
- **What is the skeleton of an organic molecule?**
- **What is a functional group?**
- **Some abbreviations used by all organic chemists**
- **Drawing organic molecules realistically in an easily understood style**